JOINT PRECISION BUDGETING ACROSS ATTENTION KV AND RECURRENT STATE IN HYBRID LINEAR-ATTENTION SERVING

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ABSTRACT

Hybrid linear-attention models pair a small set of GQA attention layers with interleaved Gated Delta Network (GDN) layers, so serving memory is split between a growing, quantizable attention KV cache and a fixed per-sequence recurrent state that no KV quantizer touches. Existing systems optimize the KV bit-width and leave state precision fixed. We show that state precision is a second, composable budget dimension. On Qwen3.5-2B/9B served with vLLM on an RTX 5090, switching the GDN state from fp32 to bf16 raises int4-KV capacity by 1.11–1.41x across context lengths 4K–16K (a conservative lower bound: all four measured cells fall below the architecture-derived model, errors -3.24% to -0.18%), and raises the compound int4-over-fp16 KV ratio from 2.245x to 2.675x at 2B/4K. Quality effects are bounded: perplexity shows no detectable change, GSM8K has a significant but under-powered 1.0-point regression at 2B and none at 9B, and RULER shows no detectable difference within wide intervals. Paired goodput gains appear in the Random60 overload region and reappear in a second formal run, while no serving cell survives multiple-comparison correction; boundaries remain workload- and threshold-dependent. We report both the capacity model and the exact statistical boundaries of every quality and serving claim.

1 INTRODUCTION

KV caches are the dominant memory consumer in LLM serving, and KV quantization is the standard lever for fitting more context and concurrency into a fixed GPU budget (Liu et al., 2024; Hooper et al., 2024; Sharma et al., 2024). The field has largely been built on attention-only transformers, in which every layer contributes a growing, homogeneous, quantizable cache.

Hybrid linear-attention models break that assumption. Qwen3.5, the subject of this paper, interleaves 18 (2B) or 24 (9B) Gated Delta Network (GDN) layers (Yang et al., 2025) with 6 (2B) or 8 (9B) GQA attention layers. The GDN layers keep no per-token KV; instead each sequence carries a fixed recurrent state that is allocated from the same GPU memory pool as the attention KV. Consequently the serving memory footprint of a hybrid model is two-dimensional: a per-token component that KV quantization can shrink, and a per-sequence state component whose size depends on its own data type. State precision is a deployment switch in vLLM (`mamba_ssm_cache_dtype`), but existing budget analyses, including prior work on state compression itself (Dao, 2026; vLLM Project, 2026), do not treat it as a dimension that should be jointly allocated with the KV bit-width.

This paper is a system study with four contributions. (a) A capacity model $C(L) = M/(A \cdot L + G)$ in which A is the per-token attention-KV footprint at the chosen KV bit-width and G is the per-sequence state footprint at the chosen state precision. All parameters are derived from the model configuration and the vLLM page layout, so the model predicts capacity at untested context lengths without probing the engine; on seven cells it matches measurement within -3.2% to $+3.5\%$, and under int4 KV all four cells fall below the prediction. (b) A capacity measurement protocol that probes exact token capacity on the serving engine and verifies the resolved configurations. (c) A quality map across perplexity, GSM8K, and RULER with paired seeds and disclosed statistical power. (d) A serving evaluation with a second formal run, in which the Random60 overload-region goodput deltas reappear and every boundary cell is reported without assigning an effect. The target regime is short-context, high-concurrency, memory-bound serving, where the fixed state is a large fraction of the budget.

The headline result is a capacity effect. Under int4 KV, bf16 state increases token capacity by 1.3755x (2B, 4K), 1.4063x (9B, 4K), 1.1149x (2B, 16K), and 1.1398x (9B, 16K), and every measured cell falls below the architecture-derived prediction, which justifies treating the model as

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a conservative lower bound for the int4 column (sign test $P = 0.0625$; errors -0.18% to -3.24%). The compound ratio of int4 over fp16 KV rises from 2.245 to 2.675 at 2B/4K when the state is also bf16. The fp16-KV column is reported cell-by-cell because its signs are mixed ($+2.86\%$, $+3.53\%$, -2.83%); block-rounding granularity shifts with layout and we do not treat the fp16 column as a bound.

The quality evidence is bounded. Perplexity changes are within $[-0.013, +0.007]$ (C4) and $[-0.045, +0.058]$ (PG19) with all confidence intervals containing zero. GSM8K (200 seeded items, 9 seeds) shows a significant 1.0-point regression at 2B state-bf16 ($p = 0.025$), which we report together with its pre-registered minimum detectable effect of 1.16 points and 67.5% observed power; the 9B model shows no regression. RULER intervals are wide (±5–33 points) and we make no equivalence claim: the protocol simply cannot detect a difference at 20 samples. Serving boundaries are workload- and threshold-dependent; paired goodput gains in the Random60 overload region reappear in a second formal run with the same contracts and seeds, but two boundary cells and the ShareGPT 250 ms direction do not, and we restrict every serving claim accordingly.

2 Background and Related Work

KV cache quantization and eviction. KIVI (Liu et al., 2024) and KVQuant (Hooper et al., 2024) established 2–4-bit per-token-head KV quantization; MiniKV (Sharma et al., 2024) and TurboQuant (Zandieh et al., 2025) push toward 2-bit regimes with system co-design. Eviction and hybrid policies (H2O (Zhang et al., 2023), StreamingLLM (Xiao et al., 2023), SnapKV (Li et al., 2024)) release memory by dropping tokens rather than reducing precision. All of these target attention-only models. On a hybrid model they compress only the attention layers; the GDN state remains untouched. Recent system-level work also co-optimizes eviction with quantization within the attention-KV budget (HqeKV (Wang et al., 2026), RDKV (Zhang et al., 2026), ARKV (Lei & Ilager, 2026)).

Hybrid linear-attention architectures. Mamba (Gu & Dao, 2023) and Mamba2 (Dao & Gu, 2024) established selective state-space layers; Gated Delta Networks (Yang et al., 2025) make the recurrence a delta rule. Production hybrids (Jamba (Lieber et al., 2024), RecurrentGemma (Botev et al., 2024), Zamba (Glorioso et al., 2024), Qwen3.5 (Team, 2026)) interleave such layers with a small number of full-attention layers. Serving them requires a per-sequence state in addition to the attention KV; vLLM's PagedAttention memory manager (Kwon et al., 2023) now allocates both from the same GPU pool.

State precision and compression. Quantizing SSM states has been explored in the low-precision inference literature: ReplaySSM proposes caching inputs instead of state, and vLLM exposes a configurable state-dtype path (Dao, 2026; vLLM Project, 2025); FP8 state checkpointing is under development in the FlashInfer SSU path (vLLM Project, 2026). Recurrent state caching also appears in inference frameworks such as FlashInfer (Ye et al., 2025). Weight-and-activation quantization of SSM layers (Quamba (Chiang et al., 2024), MambaQuant (Xu et al., 2025)) targets the compute path with W8A8 or W4A8 schemes and does not vary the per-sequence state dtype that serving allocators budget, which is the dimension we vary. QPruningKV (Zhang et al., 2025) characterized the token-precision trade-off under a fixed byte budget for attention models. State compression itself is prior art. The gap is joint budget accounting: serving systems expose state dtype without analyzing it as a precision dimension that multiplies with the KV bit-width decision, and without validating the resulting capacity-quality-serving trade-off with replicated measurements.

3 Precision Budget Model

Let L be the context length, A_f and A_q the per-token attention-KV bytes at fp16 and int4 KV respectively, and G the per-sequence state bytes at the chosen state dtype. The maximum number of context tokens served on a single GPU with M bytes available for KV plus state is

$$C(L) = \frac{L \cdot M}{A \cdot L + G}, \tag{1}$$

where $A \in \{A_f, A_q\}$ and G depends on the state dtype; the denominator is the per-sequence footprint of a length-L request, so $C(L)/L$ is the corresponding number of concurrent sequences. The capacity ratio from switching the state fp32 to bf16 is

$$r_{state}(L) = \frac{AL + G_{fp32}}{AL + G_{bf16}}, \tag{2}$$

which decreases with L: state dominates at short context, KV dominates at long context. The corresponding KV ratio at a fixed state dtype is $r_{kv}(L) = (A_f L + G)/(A_q L + G)$. The parameters A_q, A_f, and G are derived from the model configuration and the vLLM page layout (per-layer state shapes and padded block sizes), independent of the probed token counts, so the comparison of Eq. (2) against measurements is not circular. Details of the derivation and the padded block accounting are in the appendix.

4 Evaluation Setup

Platform and engine. Qwen3.5-2B and Qwen3.5-9B served from a vLLM worktree (commit `55f4768`) on a single RTX 5090 (32 GB), with `gpu_memory_utilization=0.85`. All runs record the resolved configuration (`kv_cache_dtype`,

Table 1. **Capacity ratio r_{state} from bf16 state.** Measured token capacity ratio and architecture-derived prediction; signed gap is (measured−predicted)/predicted. int4 column: all four cells are below the prediction (conservative lower bound). fp16 column reported per cell.

KV	Model	L	fp32 tok	bf16 tok	gap
int4	2B	4096	2,692,710	3,703,954	−2.37%
int4	2B	16384	4,895,837	5,458,458	−3.24%
int4	9B	4096	315,392	443,538	−0.18%
int4	9B	16384	573,440	653,635	−1.07%
fp16	2B	4096	1,199,383	1,384,448	+2.86%
fp16	2B	16384	1,552,143	1,661,337	+3.53%
fp16	9B	4096	144,104	161,899	−2.83%

`mamba_ssm_cache_dtype`, sha256 of the serving config) per attempt.

Capacity probes. We use the engine's maximum-token probe with 4K/16K context, uniform int4 KV (`int4_per_token_head`) or fp16 KV, and state fp32/bf16. Each cell is a single deterministic run: the block allocator is deterministic for a fixed engine build and configuration, and each attempt records the resolved dtype and block counts plus the sha256 of the serving configuration. The 2x2 capacity experiment is summarized in Table 1.

Quality. Perplexity is measured with a chunk-level harness (write-back rounding at chunk boundaries, chunk 128) on C4 and PG19, 5 sequences, 3 dataset seeds; we report in Section 6 why this harness is an approximation. Reasoning quality uses the vLLM kernel path: GSM8K (Cobbe et al., 2021) with 200 seeded items per seed, greedy, no chain-of-thought, 9 seeds; RULER (Hsieh et al., 2024) with 3 dataset seeds and the engine seed fixed, using the engine's default thinking mode with a 256-token generation cap (Section 6 discusses think truncation). All comparisons are paired across seeds and reported as deltas with 95% confidence intervals.

Serving. Protocol v3: piecewise constant Poisson arrivals, 120 s warmup, 60 s (300 s for boundary confirmation) measurement windows, TTFT thresholds 250–3000 ms, TPOT 200 ms, goodput/offered ≥ 0.95 required for a sustainable boundary, failures counted in the denominator, 3 seeds. Workloads: Random60 (60-token requests) and ShareGPT300 (300-token traces (LMSYS, 2023)). The formal matrix was followed by a second formal run with the same contracts and seeds; both are committed as atomic artifacts (MLSys, 2026).

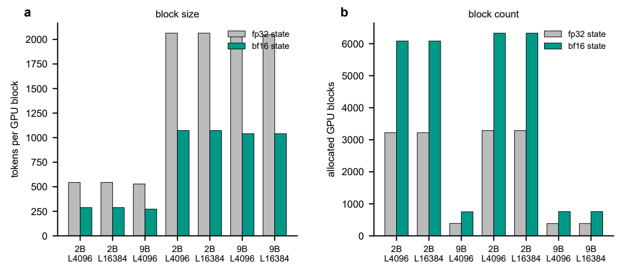

Figure 1. **Block granularity under fp32 vs. bf16 state.** (a) Tokens per GPU block and (b) number of allocated GPU blocks for the seven capacity cells. bf16 halves the per-block state footprint; the discrete re-partitioning of blocks explains the residual between measured and modeled r_{state} in Table 1.

5 Results

5.1 Capacity

Table 1 and Figure 2 report the capacity measurements. Under int4 KV, bf16 state raises capacity by 37.6% (2B, 4K), 40.6% (9B, 4K), 11.5% (2B, 16K), and 14.0% (9B, 16K). The fp16-KV column shows smaller gains (7.0–15.4% at 4K) with mixed prediction signs. The discrete page-block rounding is the source of the residual: fp32/bf16 block sizes change with layout (e.g., int4 2B: 2064/1072 tokens per block; fp16: 544/288), and the same rounding that makes the int4 column conservative flips sign under fp16 KV. We therefore frame the capacity model as a conservative lower bound only for the int4-KV column. Figure 1 shows the mechanism: bf16 state halves the per-block state footprint, which doubles the number of allocatable GPU blocks, and the measured ratio is the integer-arithmetic outcome of that re-partitioning.

Figure 2b also shows the compound KV ratio: with fp32 state, int4 KV yields 2.2451x over fp16 KV at 2B/4K; with bf16 state the same comparison yields 2.6754x. The two budget dimensions compound: the state switch and the KV switch do not share memory and their gains multiply. In deployment terms, the fp32 GDN state at 2B is 18.63 MiB per sequence (9.32 MiB at bf16). At 4K under int4 KV, the measured total-token ceilings of 2,692,710 and 3,703,954 tokens correspond to 657 and 904 concurrent length-4096 sequences on the same GPU budget, so the state switch adds roughly 247 sequence slots without changing the KV bytes themselves.

5.2 Quality: perplexity and reasoning

Perplexity. Under int4 KV, the bf16-state marginal perplexity change is -0.0029 [-0.0129, $+0.0072$] on C4 and $+0.0065$ [-0.0447, $+0.0578$] on PG19 (3 seeds, paired 95% CI). Both intervals contain zero, and the magni-

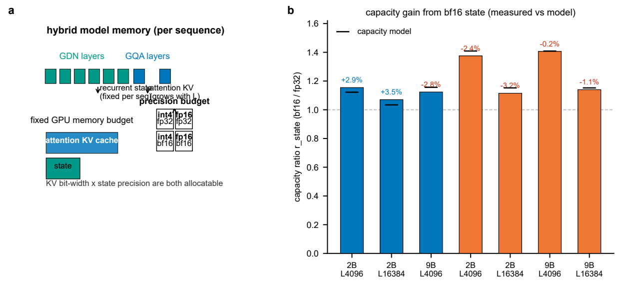

Figure 2. **Precision budget for hybrid serving.** (a) GDN recurrent state (fixed per sequence) and attention KV (growing with context) both draw from the same GPU pool; both bit-width (KV) and state precision are allocatable. (b) Measured vs. predicted r_{state} for the int4/fp16 KV columns (7 cells; gap labels are signed percent error). int4 column is a conservative lower bound: all four gaps are negative.

tude is the same as the state change under fp16 KV (-0.0003/$+0.0004$), so stacking state-bf16 onto int4 KV introduces no measurable additional perplexity cost. The harness is a chunk-level approximation (Section 6); these numbers are supporting evidence for the kernel-level claim, not a substitute for it. Figure 5 quantifies both statements: chunk-boundary rounding moves PPL by roughly 87% when the chunk size is reduced to one, while the marginal state cost under int4 KV stays within 0.007 of the fp16-KV cost on both corpora.

GSM8K. Figure 3 summarizes the 9-seed paired deltas. At 2B, bf16 state loses 1.00 point $[-1.71, -0.29]$, $p = 0.025$, with a pre-registered minimum detectable effect of 1.16 points and 67.5% observed power; this is a significant regression whose effect size is below the pre-registered MDE, and we report both. Uniform int4 KV loses 2.72 points $[-4.20, -1.24]$, $p = 0.007$, power 88.3%. Stacking state-bf16 onto int4 KV shows a marginal $+1.17$ points $[-0.33, +2.66]$ relative to int4 alone: no additional regression. At 9B, bf16 state gains 0.33 points $[-0.07, +0.73]$, $p = 0.141$: no detectable regression. We do not aggregate the 2B and 9B effects. Figure 6 shows the same data at seed resolution: the nine thin lines pair each dataset seed across allocations, and the between-seed spread is what the confidence intervals in Figure 3 summarize.

RULER. Figure 4 reports the five cells (FWE and NIAH-multiquery, 4K/8K) that the earlier KV-quantization screen had singled out; the selection predates the state-dtype grid and is not a post hoc choice from the single-seed deltas in this comparison. With three dataset seeds, every confidence interval contains zero; the FWE intervals are $\pm$5–33 points wide. We make no equivalence claim: the protocol cannot detect a difference at 20 samples per cell. The single-seed full grid (7 tasks, 2 lengths, 2 models, same protocol) is

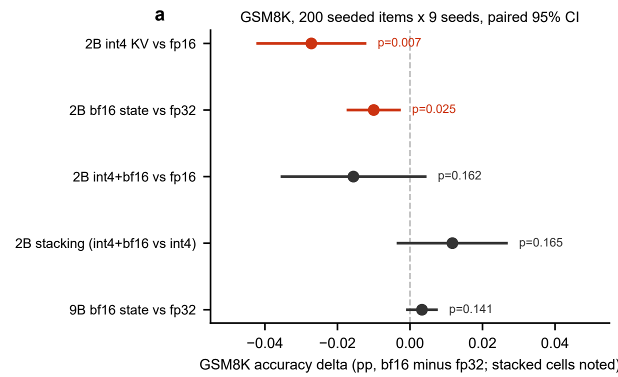

Figure 3. **GSM8K paired deltas.** 200 seeded items, greedy, no chain-of-thought, 9 dataset seeds; points and 95% paired CIs. Red intervals exclude zero. The 2B state and int4 regressions are significant; the 2B state effect is below the pre-registered MDE (1.16 pp) with 67.5% power.

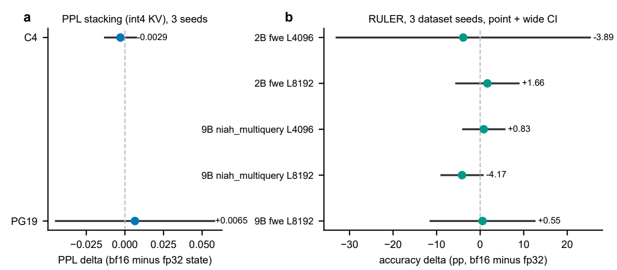

Figure 4. **Perplexity stacking and RULER.** (a) PPL marginal change of bf16 state under int4 KV (C4/PG19, 3 seeds). (b) RULER deltas (bf16 minus fp32 state) for the five screen-selected cells (FWE and NIAH-multiquery, 4K/8K); 3 dataset seeds, default thinking mode (max 256 tokens), wide CIs, no detectable difference.

archived and summarized in Appendix Table 3: mean deltas are $+0.49$ points (2B) and -0.71 points (9B); apart from the five re-tested cells, the nonzero single-seed cells lie within ±1.7 points.

5.3 Per-layer sensitivity

We separately asked whether any single GDN layer should be kept at higher precision. Bf16 was applied to one GDN layer at a time (18 configurations, 2 corpora, 3 seeds, 36 tests). Two raw intervals excluded zero (C4, layers 2 and 8, $p = 0.049$ and $p = 0.0036$), but neither survives Bonferroni ($\alpha/36$) or BH-FDR correction. There is no evidence for a per-layer precision benefit, so the paper's allocation dimension is the whole-state switch rather than layer-wise state precision (Figure 7).

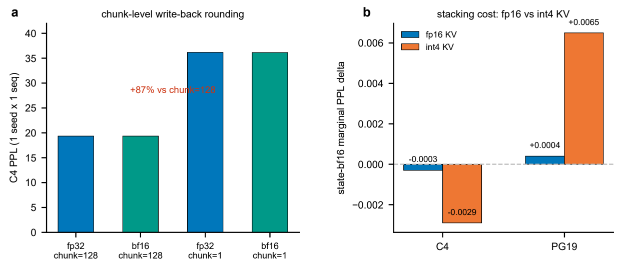

Figure 5. **Harness boundary and stacking cost.** (a) Chunk-level write-back rounding: chunk=1 raises C4 PPL by roughly 87% relative to chunk=128 (1 seed, 1 sequence), so PPL is a chunk-level approximation. (b) Marginal state-bf16 PPL delta under fp16 KV vs. int4 KV (3 seeds): stacking introduces no measurable additional cost.

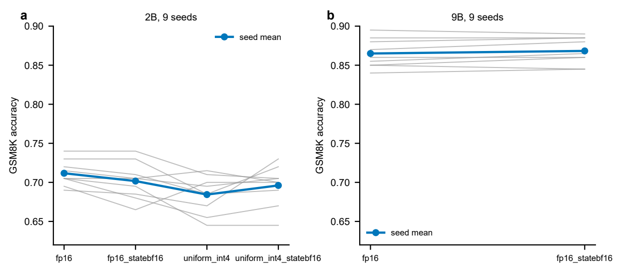

Figure 6. **GSM8K per-seed paired accuracy.** 200 seeded items, greedy, no chain-of-thought, 9 dataset seeds. Thin lines connect the same seed across allocations; the blue line and markers are seed means. (a) 2B across four allocations; (b) 9B across two.

5.4 Serving

Table 2 reports sustainable arrival rates for `int4` and `int4+bf16` allocations under the v3 protocol, and Figure 8 shows paired goodput deltas in the overload region.

The only repeated boundary difference is Random60 at 1000 ms (35 vs. 40 in the formal matrix), and it did not reappear in the second run (int4 also reached 40 there). ShareGPT at 250 ms shows opposite directions between the two runs. We therefore do not present either workload as improved; the boundary tables are measurements, and we report them without assigning an effect.

What does replicate is the paired goodput delta in the Random60 overload region. At r40, bf16 state improves goodput by 0.334 [0.078, 0.589] at 250 ms (paired t-test $p = 0.030$) and 0.215 [0.154, 0.276] at 500 ms ($p = 0.004$); the second formal run measures 0.304 [0.250, 0.359] ($p = 0.002$) and 0.138 [0.082, 0.195] ($p = 0.009$). At r45/r50 and 2000–3000 ms the deltas are smaller but consistently positive in both runs (e.g., r45/3000 ms: 0.367 [0.308, 0.426] vs. 0.372 [0.310, 0.435]; $p = 0.001$ and $p = 0.002$). None of the 60

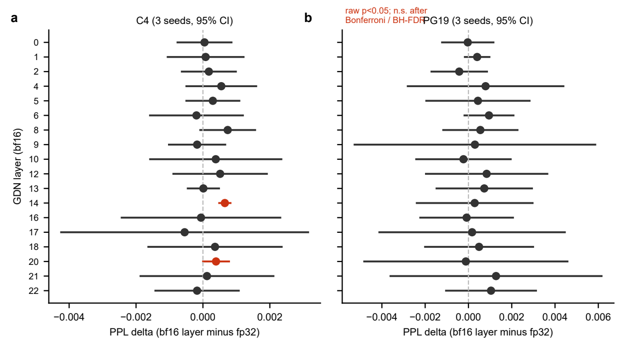

Figure 7. **Per-layer state sensitivity (2B, C4/PG19, 3 seeds).** Each row is bf16 applied to one GDN layer; points and paired 95% CIs are deltas against the fp32 baseline. Red rows (C4 layers 2 and 8) have raw $p < 0.05$ and are not significant after Bonferroni or BH-FDR correction.

Table 2. **Sustainable rate boundaries (req/s, 3 seeds, protocol v3).** Formal matrix; values in parentheses are the second formal run (identical contracts and seeds). Differences between runs at Random60/1000 ms and ShareGPT/250 ms show that single boundary cells are run-sensitive and are not claimed as effects.

Workload	Config	250	500	1000	2000	3000
Random60	int4	30	35	35 (40)	40	40
Random60	int4+bf16	30	35	40	40	40
ShareGPT	int4	40 (35)	40	40	40	40
ShareGPT	int4+bf16	35 (40)	40	40	40	40

serving cells survives BH-FDR at $q < 0.05$ (closest: formal r40/500 ms $q = 0.066$; second run r40/250 ms $q = 0.052$). We therefore treat the overload-region deltas as directional evidence whose weight comes from same-cell replication across the two runs, and we stop short of treating them as sustainable-SLO improvements. Per-cell p and BH q values for all 60 cells are archived with the analysis artifacts. We report the goodput deltas as aggregate effects and do not assign them to a mechanism; Section 6 states the confounded channels and the discriminator we leave for future work. ShareGPT at 500 ms and above shows boundary parity in both runs.

6 Limitations

Harness boundary. The perplexity harness rounds the state at chunk boundaries (chunk 128). A chunk-1 ablation changes PPL from 19.35 to 36.16 (+87%), which shows the harness is not equivalent to per-token kernel semantics. We treat PPL as supporting evidence and place the kernel-path quality conclusions (GSM8K, RULER) on the vLLM serving path.

RULER protocol. RULER runs used the engine's default

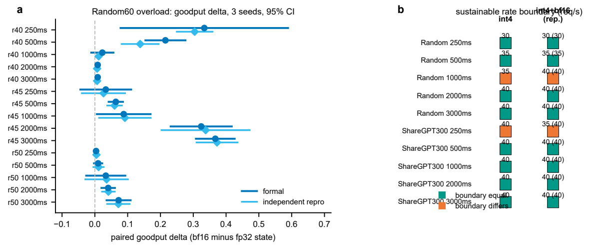

Figure 8. **Serving, Random60 overload region.** (a) Paired goodput delta (bf16 minus fp32 state) at rates 40–50 req/s, all thresholds, formal and second run, 3 seeds each, 95% CI. (b) Sustainable-rate boundary parity map (int4 vs. int4+bf16); orange cells differ between allocations, teal cells are equal, with the second-run value in parentheses.

thinking mode with a 256-token generation cap. FWE scores can be sensitive to think truncation, so the wide intervals in Figure 4 may shift in either direction under a no-think protocol; a no-think re-run of the five cells is future work.

Generalization. All evidence is from Qwen3.5-2B/9B on one RTX 5090. Claims are scoped to GDN-based hybrids; Mamba2-style architectures and other hardware are untested. fp16-state quality is smoke-tested only; fp8/int8 state precisions are future work, so we make no claims about a full precision spectrum.

Serving variability. Overload-region goodput is stable in direction but not in magnitude across runs; boundary cells are run-sensitive. A workload with very long requests or a different mix may not show the effect. We did not test offloading or prefix caching, each of which changes the memory equation.

Mechanism attribution. The serving deltas are aggregate effects: bf16 state changes state bytes, tokens per block, and the number of allocatable blocks at once, so capacity, bandwidth, and allocator granularity are confounded. A fixed-block-count contrast (same number of GPU blocks while varying state bytes) or a fixed-bytes contrast (same state bytes while varying block geometry) would separate these channels; neither was run. The mechanism behind the serving benefit remains open, and we claim only the reproduced aggregate delta.

Tensor parallelism. The single-GPU ratios extend to tensor parallelism only under uniform sharding. If the GDN state is sharded by its hidden dimension and the attention KV by its KV heads, both $A \cdot L$ and G scale as $1/P$, so $r_{state}(L)$ is unchanged to first order. Per-rank block rounding and communication overhead can shift the measured ratio, and we did not test TP=2/4; the transfer is an expectation, not a measurement.

Statistical resolution. GSM8K at 2B detects a 1.0-point regression with 67.5% power; effects smaller than $\sim$1.16 points are below the pre-registered MDE. RULER cannot distinguish differences smaller than tens of points at 20 samples.

7 Conclusion

Hybrid linear-attention serving has two allocatable precision dimensions: the attention-KV bit-width and the recurrent-state dtype. We measured the capacity, quality, and serving consequences of treating both as a joint budget. The capacity model is a conservative lower bound under int4 KV, the compound KV $\times$ state gain is real (2.245 to 2.675 at 2B/4K), quality degradation is bounded and disclosed at the level of statistical power, and the serving benefit is a workload-limited paired goodput effect reproduced in a second formal run. We release the full measurement contracts, per-seed data, and reproduction artifacts so that the boundary cells and null results can be audited independently. Long-form and retrieval evidence beyond GSM8K and RULER (e.g., LongBench (Bai et al., 2024)) remains single-seed pilot evidence and is excluded from the claims above.

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A Capacity model derivation

For Qwen3.5-2B, the GDN state per sequence is derived from the vLLM `MambaStateShapeCalculator` path: 18 layers $\times$ 1,085,440 bytes per layer at fp32 (code-derived; the 9B state is 24.84 MiB). The per-token attention-KV footprints A_f and A_q are computed from the GQA head/layer configuration and the vLLM packed page layout (fp16: 2 bytes per element; int4: 0.5 bytes per element). G for bf16 is half of G_{fp32}. Eq. (2) uses these derived values; the predicted ratios reported in Table 1 are evaluated with no fitted parameters. The observed residual is attributable to the allocator's discrete block rounding, which we verified from the per-attempt block-size and block-count records.

For Eq. (1), $C(L)/L$ is the number of concurrent length-L sequences.

Table 3. **Single-seed RULER grid.** bf16-minus-fp32 state accuracy deltas (percentage points) for the full 7-task $\times$ 2-length grid, protocol as in Section 4. One dataset seed with the engine seed fixed; values are a screen, not a multiplicity-corrected test. The five cells with 3-seed re-tests (Figure 4) were fixed by the earlier KV-quantization screen. Overall mean deltas are $+0.49$ points (2B) and -0.71 points (9B).

Model	Task	$L = 4096$	$L = 8192$
2B	NIAH single	0.00	0.00
2B	NIAH multikey	0.00	0.00
2B	NIAH multivalue	0.00	-1.25
2B	NIAH multiquery	0.00	0.00
2B	VT	0.00	$+1.00$
2B	CWE	0.00	$+0.50$
2B	FWE	$+8.33$	-1.67
9B	NIAH single	0.00	0.00
9B	NIAH multikey	0.00	0.00
9B	NIAH multivalue	0.00	0.00
9B	NIAH multiquery	$+1.25$	-6.25
9B	VT	0.00	0.00
9B	CWE	0.00	0.00
9B	FWE	0.00	-5.00